Som Sagar

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EDUCATION

Arizona State University — Tempe, Arizona
*PhD Candidate, Computer Science – **GPA:** 3.9/4* — August 2023 – December 2027 (Expected)

SELECTED PUBLICATIONS

- **S. Sagar*** J. Duan, S. Vasudevan, H. Ben'Amor, D. Fox **"RoboMD: Uncovering Robot Vulnerabilities through Semantic Potential Fields"** International Conference on Learning Representations **(ICLR), 2026**. [paper]
- Atharva G* **S. Sagar*** R. Senanayake, **"PAC Bench: Do Foundation Models Understand Prerequisites for Executing Manipulation Policies?,"** Neural Information Processing Systems **(NeurIPS), 2025**. [paper]
- A. Taparia, **S. Sagar**, R. Senanayake, **"Explainable Concept Generation through Vision-Language Preference Learning"** International Conference on Machine Learning **(ICML), 2025**. [paper]
- **S. Sagar**, A. Taparia, H. Mankodiya, P. Bidare, Y. Zhou, R. Senanayake, **"Trustworthy Explanations for Robot Behaviors,"** International Conference on Intelligent Robots and Systems **(IROS), 2025**. [paper] *(Oral)*
- **S. Sagar**, A. Taparia, R. Senanayake, **"Failures Are Fated, But Can Be Faded: Characterizing and Mitigating Unwanted Behaviors in Large-Scale Vision and Language Models,"** International Conference on Machine Learning **(ICML), 2024**. [paper] *(Spotlight)*

WORK EXPERIENCE

Google LLC — Sunnyvale, California
Student Researcher — May 2026 – August 2026

- Built a per-query **agentic workflow router** that represents multi-agent systems as a **Directed Acyclic Graph** and compiles a custom topology via **Monte Carlo Tree Search** (20 simulations/state), lifting **GAIA** 14%→69% and **SpreadsheetBench** 46%→81% over zero-shot, beating LATS, TRACE, GPTSwarm, and DyLAN.
- Adapted a frozen **Gemma (2B)** backbone by swapping its 250K-token vocabulary head for 10-action **policy** and scalar **value heads** (6.8M **LoRA** parameters), warm-started from PASS−FAIL embedding arithmetic.
- Trained via **policy-value distillation** on MCTS visit-count targets under a **token- and latency-aware composite reward**, cutting policy loss $\sim$10$\times$ and converging to compact 4-node workflows instead of agent sprawl.

LinkedIn Corporation — Sunnyvale, California
Machine Learning Research Intern — May 2025 – August 2025

- Designed a **multi-head preference** RL framework where each head learns a distinct reasoning pattern and user preference signal rather than collapsing them into a single reward, lifting decision quality by 20%.
- Engineered structured reward pipelines driven by **human-in-the-loop** feedback, collecting and curating 1,200+ annotations to train and validate the reward model, raising **agent alignment** success rates by 40%.
- Integrated **HydraLoRA** adapters into the agent training loop to enable **parameter-efficient fine-tuning** across multiple task domains without retraining the base model, cutting compute costs by 25%.

RELEVANT PROJECTS

LENS Lab, *Reinforcement Learning, Large Language Models* — August 2023 – Present

- Developed a **Proximal Policy Optimization** (PPO) method to surface **out-of-distribution** (OOD) instances in classifiers and language models (T5, BART); **fine-tuning** cut OOD errors by 30% and raised accuracy by 13%.
- Reduced bias in the Stable Diffusion v1-5 model by 37% through **LoRA** fine-tuning while improving output quality.
- Introduced a DQN-based concept generation framework using **TCAV** (Testing with Concept Activation Vectors) rewards and a **preference-optimizing** step function for concept-based explanations.
- Formulated a **Bayesian TCAV** framework with **uncertainty estimation**, improving interpretability of robotic actions by 22% across MuJoCo, DonkeyGym, and JetBot.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++
- **Technologies/Frameworks:** PyTorch, TensorFlow, Transformers, TRL, PEFT/LoRA, Diffusers, verl, Stable Baselines, Gymnasium, Captum, Scikit-learn, NumPy, Pandas, Google Cloud Platform (GCP)

AWARDS AND ACHIEVEMENTS

- SCAI Conference Award, School of Computing and Augmented Intelligence — May 2025
- ICML 2024 Spotlight Paper — Top 3.5% of submissions — July 2024